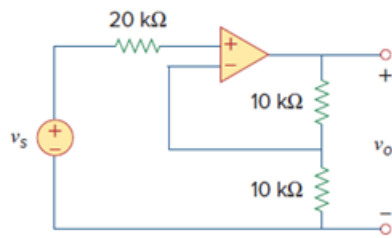


## Problem

Find the gain  $v_o/v_s$  of the circuit in Fig. 5.49.

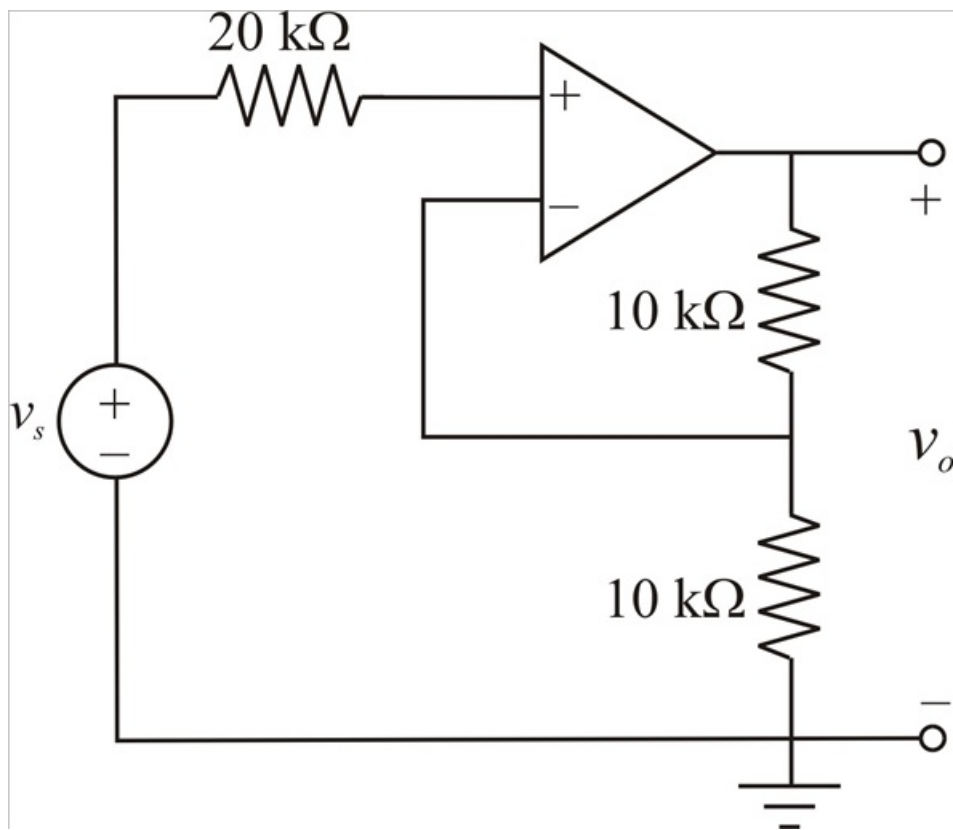
**Figure. 5.49:**



## Step-by-step solution

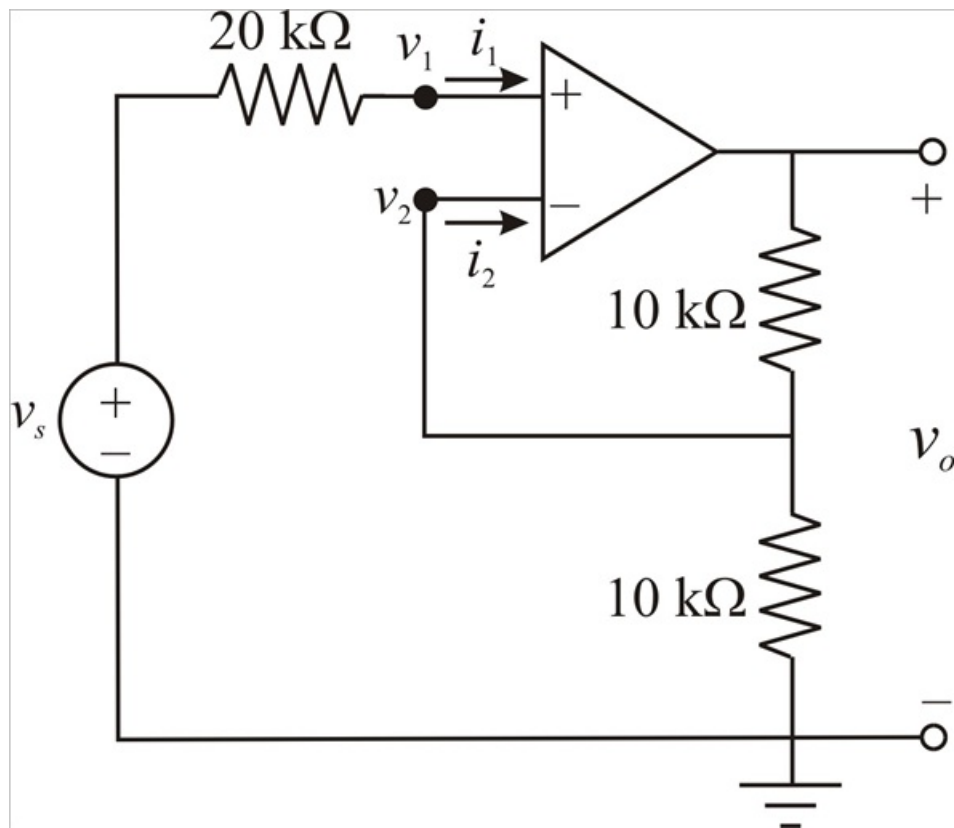
## Step 1 of 5

Given circuit diagram is:



## Step 2 of 5

Above circuit can be represented as



### Step 3 of 5

In an ideal op-map, the inverting and non-inverting terminal currents are zero. And the inverting and non-inverting node voltages are equal.

That is,

$$i_1 = i_2 = 0 \text{ A and}$$

$$v_1 = v_2$$

### Step 4 of 5

Applying Kirchhoff's current law at node  $v_1$ :

$$\frac{v_1 - v_s}{20 \times 10^3} + i_1 = 0$$

$$\frac{v_1 - v_s}{20 \times 10^3} + 0 = 0 \quad (\text{Since, } i_1 = i_2 = 0 \text{ A})$$

$$\frac{v_1 - v_s}{20 \times 10^3} = 0$$

$$v_1 - v_s = 0$$

$$v_1 = v_s$$

Hence, we get  $v_1 = v_s$ .

### Step 5 of 5

Applying Kirchhoff's current law at node  $v_2$ :

$$\frac{v_2}{10 \times 10^3} + \frac{v_2 - v_o}{10 \times 10^3} + i_1 = 0$$

$$\frac{v_2}{10 \times 10^3} + \frac{v_2 - v_o}{10 \times 10^3} + 0 = 0 \quad (\text{Since, } i_1 = i_2 = 0 \text{ A})$$

$$v_2 + v_2 - v_o = 0$$

$$2v_2 - v_o = 0$$

$$2v_1 - v_o = 0 \quad (\text{Since, } v_1 = v_2)$$

Substitute  $v_1 = v_s$  in the above equation

$$2v_s - v_o = 0$$

$$v_o = 2v_s$$

$$\frac{v_o}{v_s} = 2$$

Therefore, the voltage gain is

$$\boxed{\frac{v_o}{v_s} = 2 \text{ V/V}}$$